



2.7

Composition of Functions

Student Notes and Guided Practice

AP Precalculus - Unit 2

| ECHS Mathematics

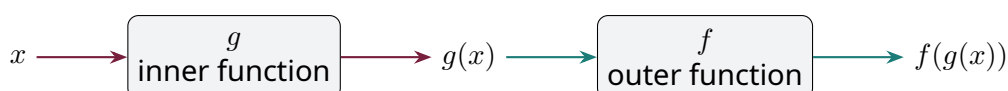
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Learning Targets

- Evaluate compositions from formulas, tables, graphs, and contexts.
- Determine domains of composite functions.
- Construct and decompose composite functions.

Essential Question

How does the output of one function become the input of another?



Concept Framework

Key Idea

Composition order matters: generally $f \circ g \neq g \circ f$.

Key Idea

The domain of $f \circ g$ consists of inputs in the domain of g whose outputs lie in the domain of f .

Key Idea

Units can reveal meaningful composition order.

Key Idea

Decomposition is not always unique.

Formula Map

Formula and Structure

$$(f \circ g)(x) = f(g(x))$$

Formula and Structure

$$\text{Dom}(f \circ g) = \{x \in \text{Dom}(g) : g(x) \in \text{Dom}(f)\}$$

Worked Examples

Worked Example 1

Problem. Let $f(x) = \sqrt{x}$ and $g(x) = x - 5$. Find $f \circ g$ and its domain.

Solution. $\sqrt{x - 5}$, domain $[5, \infty)$.

Worked Example 2

Problem. For $f(x) = 2x + 1$, $g(x) = x^2$, compare $f \circ g$ and $g \circ f$.

Solution. $2x^2 + 1$ versus $(2x + 1)^2$; they are not equal.

Worked Example 3

Problem. Decompose $h(x) = \ln(3x^2 + 1)$.

Solution. One choice is $g(x) = 3x^2 + 1$, $f(u) = \ln u$.

Multiple-Representation Connection

AP Reasoning

For this topic, always connect at least two representations: algebraic structure, numerical evidence, graphical behavior, or contextual meaning. State restrictions and units before interpreting a result. A correct calculation without a valid domain or contextual interpretation is incomplete AP reasoning.

Common Errors to Avoid

Common Error Check

- Do not discard domain restrictions when rewriting an expression.
- Do not infer local behavior from only one interval-wide average.
- Do not report a numerical answer without units or contextual meaning when quantities are named.
- Do not use a graphing window as proof that a feature does not exist.

Guided Check

1. Given $f(x) = 2x + 3$, $g(x) = x^2$, compute the indicated natural compositions and domains.

2. Given $f(x) = \sqrt{x}$, $g(x) = x - 4$, compute the indicated natural compositions and domains.

3. Given $f(x) = 1/x$, $g(x) = x + 2$, compute the indicated natural compositions and domains.

4. Given $f(x) = \ln x$, $g(x) = x^2 + 1$, compute the indicated natural compositions and domains.

5. Given $f(x) = x^3$, $g(x) = \sqrt[3]{x-1}$, compute the indicated natural compositions and domains.

6. Find domain of $\sqrt{\frac{1}{x-2}}$. Interpret as a composition.

Lesson Summary

Complete these sentences before leaving the lesson:

1. The most important structure in this topic is _____.
2. A restriction I must remember is _____.
3. One graphical feature I can justify algebraically is _____.